SAMARTH

CUET PG 2027

Lecture - 05 | Computer Science

Unit 03 : Digital Fundamental

Topic : Fixed Point Representation

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Recap *of previous lecture*

1. Floating Point Representation



Topics *to be covered*

1. Fixed Point Representation



Topic : Fixed Point Representation

Q Expected Type of Q

The value of a float type variable is represented in Sigle precision 32 bit floating pt IEEE 754.

A float variable X is assigned the decimal value −14.25.

X in hexadecimal notation is ——

① C164 0000 H
② 416C 0000 H
③ 4164 0000 H
④ C16C 0000 H

C = 1100
4 = 0100

Sigle precision

1 bit	8 bits	23 bits.
Sign	Exponent	Mantissa.
1		

Trick:- Negative ⇒ sign bit 1

$(14.25)_{10} = (?)_2$

2	14
2	7 0
2	3 1
	1 1

1110.010

$.25 \times 2 = 0.50$

$.50 \times 2 = 1.00$

$.00 \times 2 = 0$

Rule:- Move . to the right of 1st One

$1.11001\underline{0} \times 2^{3}$

Mantissa.

Sigle precision $(1.N)2^{E-127}$

Exponent $E - 127 = 3$

$E = 3 + 127$

$= (130)$

$= (10000010)_2$

Sign	Exponent	Mantissa
1	10000010	11001000000000000000000

1100 0001 0110 0100 0000 0000

C 1 6 4

Topic : Fixed Point Representation

Decimal to Binary

$(\quad)_{10}$ to $(\quad)_2$

Number System

Topic : Fixed Point Representation

Q. -40.1

① $(40.1)_{10} = (?)_2$

$(101000.\underline{0001}\,\underline{1001}\,1)$

Negative no sign = 1

Positive no sign = 0

② Normalize

Rule.- Move . to Right of 1st one

$1.01\,\underline{0000}\,\underline{0011}\,\underline{0011} \times 2^5$

Mantissa

③ $E-127 = 5$

$E = 5+127$

$= (132)$

Sign	Exponent	Mantissa
1	1000 0100	0100 0000 1100 1100

$1100\,0010\,0100\,0000\,0110\,0110 = (10000100)_2$

C 2 2 0 6 6

① C22066

② C20066

③ C22060

④ C22666

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Q:- Convert a floating pt no to Decimal

Given :-

0 1 1 0 1 0 0 0 0 0 0 0 0 0 0 1 1

Sign | Mantissa | Exponent

Solution :- Exponent

$(00001\,1)_2 = (3)_{10}$

$= 2^{(3)}$

$0\,1\,1\,0\,1\,0\,0\,0\,0\,0 \times 2^3$

↓ Decimal

0110 | .100000

$(6)_{10}$ | $\times 2^{-1} = \frac{1}{2} = 0.5$

$(0.5)_{10}$

$(6.5)_{10}$ Ans.

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$.10$

$\times 2^{-1} = \frac{1}{2} = 0.5$

$\times 2^{-2} = 0$

0.5

Topic : Fixed Point Representation

Number System is base

Topic : Fixed Point Representation

Fixed pt

16.25

Homework from SAMARTH CUET PG 2027 Module

Number System

Conversions

Revise

THANK YOU